

GEORGE CHEMMALA

404-932-9886 | george_chemmala@brown.edu | george.chemmala.com | github.com/Geoc2022

EDUCATION

Brown University

Providence, RI

Sc.B. Computer Science, Sc.B. Mathematics (Major GPA: 4.0)

May 2026

- **Relevant Courses:** Machine Learning | Deep Learning | Robust Algorithms for ML | Compilers | Systems | Formal Proof and Verification | Applied & Pure Cryptography | Abstract Algebra | Number Theory | AI & Euclidean Geometry

EXPERIENCE

Research Assistant, Transformer Pruning | Brown Department of CS & Physics

Summer 2025 - Present

- Implemented a novel transformer pruning method on the ParT particle jet classifier to be used on FPGAs for the CMS Trigger at CERN, resulting in an expected 30% reduction in model size and improved inference speed
- Surveyed state-of-the-art transformer pruning techniques and utilized advanced PyTorch features, including model hooks, to design and integrate custom pruning logic that utilized GPU support
- Integrated pruning pipeline into reproducible workflows emphasizing determinism and verification-friendly design

Head Teaching Assistant, Machine Learning | Brown University

Spring 2024 - Present

- Machine Learning (Spring 2026), Theory of Computation, Applied ODEs, Stats Inference, Abstract Algebra
- Lead course logistics, coordinate TA team, run problem-solving sessions, hold office hours, for proof-heavy courses

Software Engineer Intern | Aiguard

Summer 2025

- Deployed a scalable full-stack web platform for solar installation prediction and analysis, integrating FastAPI and PostgreSQL for robust backend APIs working with Google Cloud Platform and React/TypeScript for a responsive frontend
- Developed an efficient Boosted Random Forest Classifier (86% accuracy) and Forecaster for actionable insights for the client
- Added model interpretability using SHAP/LIME (AI explanation tools), building visual dashboards for stakeholders
- Refactored backend codebase for modularity and extensibility, applying OOP principles and reducing technical debt, resulting in a 2x reduction in code size and improved maintainability

PROJECTS/RESEARCH

AlphaGeometry: Neuro-Symbolic Geometry Solver | Python, Rust, Ascent, Gemini

Fall 2025

- Implemented a neuro-symbolic geometry reasoning system inspired by Google DeepMind's AlphaGeometry
- Developed a deductive database and algebraic reasoning pipeline that integrated with a LLM to propose constructions
- Extended the system to work with a relational database backend for improved efficiency and extensibility
- Continuing work with Google researchers to build systems for other domains with the relational database extension

WeensyOS: Virtual Memory Kernel Implementation | C, C++, x86 Assembly

Spring 2025

- Engineered kernel code to manage user/kernel virtual memory and enforce correct access permissions
- Enhanced memory utilization by 3x, dynamically allocating virtual pages to non-contiguous physical memory
- Optimized fork and exit system calls for efficient process creation/exit with shared memory
- Incorporated process isolation by overlapping virtual address spaces, enabling full process virtual space utilization

Vote: Cryptographic Voting Protocol Implementation | C++, CryptoPP

Spring 2025

- Designed and implemented a cryptographic protocol inspired by Helios with formal correctness and security guarantees
- Integrated zero-knowledge proofs and blinding to protect voter anonymity and ensure ballot integrity
- Extended the protocol to support multiple candidates and a restriction on the number of votes per user using a ZKP

Robust Estimation for the Erdős-Rényi Model | Python, NumPy, NetworkX

Fall 2024

- Awarded Top Project in grad course CS2952Q (Robust Algorithms for ML) Final Projects, Fall 2024
- Developed and implemented robust algorithms to estimate the edge probability p in Erdős-Rényi graphs under adversarial edge perturbations for adversarial perturbation $\varepsilon < 1/2$
- Proved partial theoretical guarantees on runtime and estimation accuracy, outperforming existing methods
- Analyzed empirical evaluations on synthetic graph data (+10k nodes), demonstrating real world improvements

SKILLS & INTERESTS

Languages: Python, C, C++, OCaml, Java, JavaScript, Lean, Isabelle, Haskell, Bash, Julia, LaTeX, Mathematica, MATLAB

Developer Tools: VS Code, Vim, PyCharm, IntelliJ, Google Colab, Jupyter, Git, Unix, Linux, Docker, Figma

Interests: Education, Music, Board/Card Games, Photography, Manim